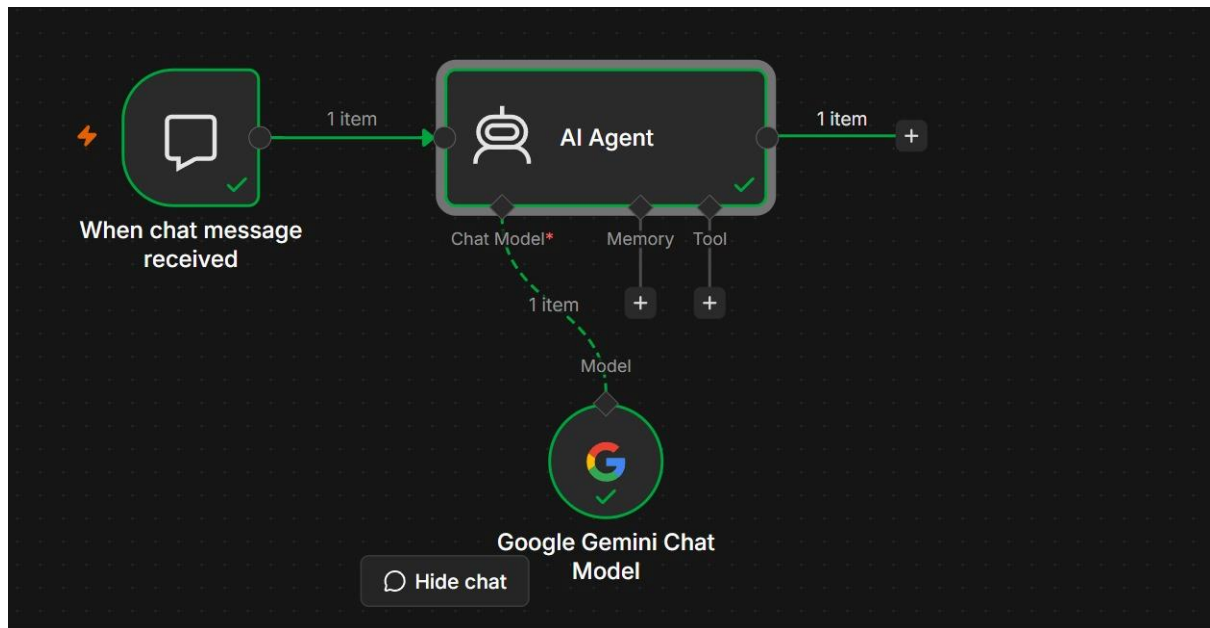


PROJECT-21

AI Recipe Generator



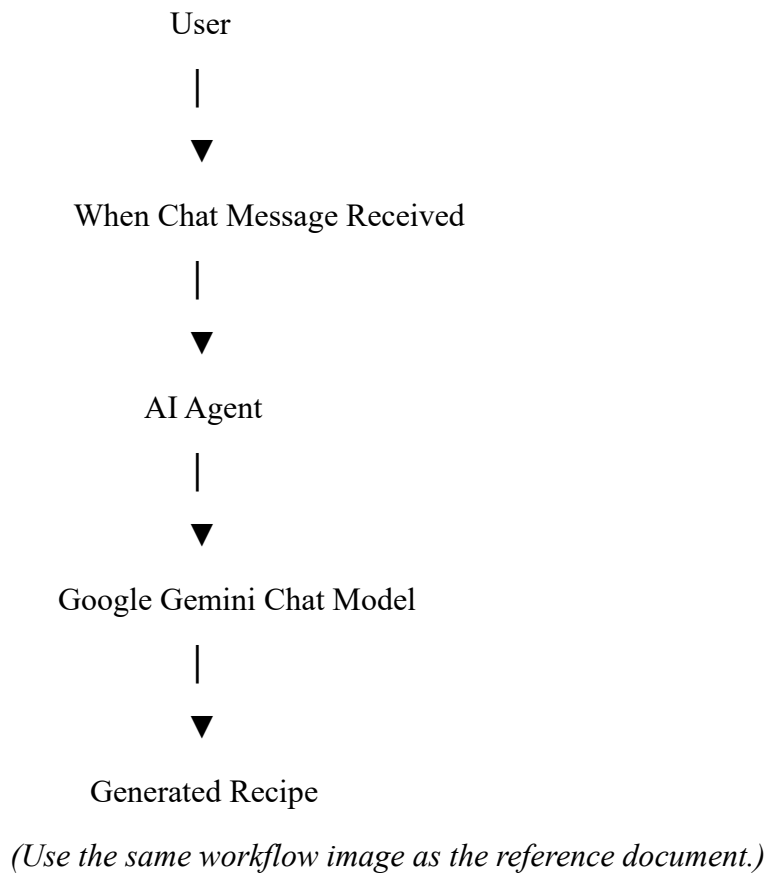
Project Objective

The objective of this project is to build an **AI-powered Recipe Generator** that creates delicious, personalized, and easy-to-follow recipes based on the user's ingredients, cuisine preferences, dietary restrictions, or meal type.

The AI Agent can generate:

- Breakfast recipes
- Lunch recipes
- Dinner recipes
- Dessert recipes
- Vegetarian recipes
- Vegan recipes
- High-protein recipes
- Keto recipes
- Indian recipes
- Healthy meal plans

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's recipe request
2	AI Agent	Understands the user's ingredients and cooking preferences
3	Google Gemini Chat Model	Generates a personalized recipe

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
- Google AI Studio API Key
- Google Gemini Chat Model configured
- Internet connection

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI Recipe Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's recipe request.

Example Input

Create a healthy vegetarian dinner recipe using paneer, spinach, tomatoes, and onions.

Output

```
{  
  "chatInput": "Create a healthy vegetarian dinner recipe using paneer, spinach, tomatoes, and onions."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Recipe Generator.

Your task is to create delicious, easy-to-follow, and personalized recipes based on the user's request.

Instructions:

- Understand the user's ingredients, cuisine preference, dietary restrictions, and meal type.
 - Generate a suitable recipe using the provided ingredients whenever possible.
 - Create a creative recipe title.
 - Mention:
 - Preparation Time
 - Cooking Time
 - Servings
 - List all required ingredients with quantities.
 - Provide clear, step-by-step cooking instructions.
 - Suggest optional ingredient substitutions when appropriate.
 - Include useful cooking tips or serving suggestions.
 - Ensure the recipe is easy for beginners to follow.
 - Return only the completed recipe.
-

Step 5 – Add Google Gemini Chat Model

Click

+ Chat Model

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)
Authentication	Google AI Studio API Key

Connect it to the **AI Agent**.

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Create a healthy vegetarian dinner recipe using paneer and spinach.



Step 2

The **When Chat Message Received** node receives the user's request.



Step 3

The **AI Agent** analyzes the request, identifies the ingredients, cuisine preference, and dietary requirements, and prepares the prompt.

↓

Step 4

The prompt is sent to the **Google Gemini Chat Model**.

↓

Step 5

Gemini generates a complete recipe including the title, ingredients, preparation time, cooking time, servings, cooking instructions, and helpful tips.

↓

Step 6

The personalized recipe is returned to the user.

Sample Inputs and Outputs

Example 1

Input

Create a pasta recipe using tomatoes, garlic, and basil.

Output

Recipe: Garlic Tomato Basil Pasta

Preparation Time: 10 minutes

Cooking Time: 20 minutes

Servings: 2

Ingredients

- 200 g pasta
- 2 tomatoes
- 3 garlic cloves
- Fresh basil
- Olive oil
- Salt
- Black pepper

Instructions

1. Cook the pasta until al dente.
 2. Heat olive oil and sauté garlic.
 3. Add chopped tomatoes and cook until soft.
 4. Mix in basil, salt, and pepper.
 5. Toss the pasta with the sauce and serve hot.
-

Example 2

Input

Generate a healthy breakfast recipe.

Output

Recipe: Oats and Fruit Breakfast Bowl

Preparation Time: 10 minutes

Cooking Time: 5 minutes

Servings: 1

Ingredients

- Oats
- Milk
- Banana
- Blueberries
- Honey
- Chia seeds

Instructions

1. Cook oats with milk.
 2. Add sliced banana and blueberries.
 3. Drizzle honey on top.
 4. Sprinkle chia seeds and serve.
-

Example 3

Input

Create a vegetarian Indian dinner recipe.

Output

Recipe: Paneer Butter Masala

Preparation Time: 15 minutes

Cooking Time: 30 minutes

Servings: 4

Ingredients

- Paneer
- Tomatoes
- Onions
- Butter
- Fresh cream
- Ginger-garlic paste
- Indian spices

Instructions

1. Prepare the tomato-onion gravy.
 2. Add ginger-garlic paste and spices.
 3. Stir in butter and cream.
 4. Add paneer cubes.
 5. Simmer for 10 minutes and serve with naan or rice.
-

Example 4

Input

Suggest a high-protein snack recipe.

Output

Recipe: Greek Yogurt Protein Bowl

Preparation Time: 5 minutes

Cooking Time: 0 minutes

Servings: 1

Ingredients

- Greek yogurt
- Mixed nuts
- Berries
- Honey
- Chia seeds

Instructions

1. Add yogurt to a serving bowl.
2. Top with berries and nuts.
3. Drizzle honey.
4. Sprinkle chia seeds and serve chilled.

Workflow Diagram

User



When Chat Message Received



AI Agent



Google Gemini Chat Model



Generated Recipe

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the **Prompt** field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Recipe Generator in n8n.
 - Configure an AI Agent with Google Gemini.
 - Design effective prompts for recipe generation.
 - Generate personalized recipes using AI.
 - Build conversational AI workflows for cooking assistance.
-

Real-World Applications

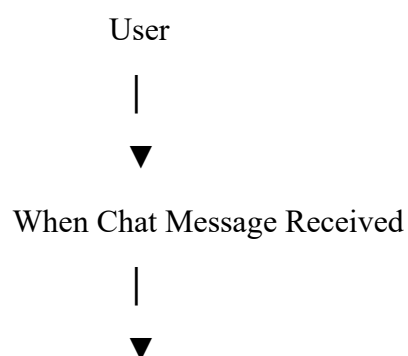
- Smart cooking assistants
 - Personalized meal planning
 - Healthy diet recommendations
 - Restaurant menu inspiration
 - Food blogging
 - Cooking education
 - Nutrition guidance
 - Recipe recommendation systems
 - Home kitchen automation
 - AI-powered food applications
-

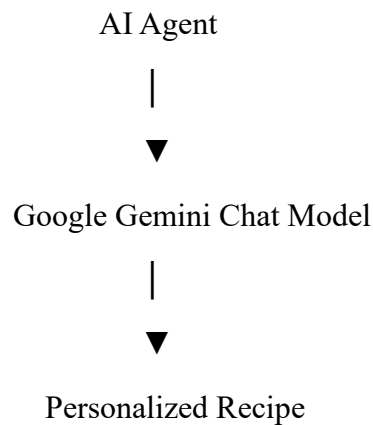
Practice Challenge

Enhance this project by adding:

1. A cuisine selector (Indian, Italian, Chinese, Mexican, etc.).
 2. A dietary preference selector (Vegetarian, Vegan, Keto, Gluten-Free).
 3. A cooking time selector (15, 30, or 60 minutes).
 4. A Memory node to remember the user's favorite cuisines and dietary preferences.
 5. A Google Sheets node to save generated recipes.
 6. An HTTP Request node or image generation service to generate an image of the finished dish automatically.
-

Final Workflow





Expected Result

The **AI Recipe Generator** accepts ingredients, cuisine preferences, dietary restrictions, or meal requirements from the user, uses the **Google Gemini Chat Model** to generate a complete recipe with preparation time, cooking time, servings, ingredients, step-by-step instructions, and cooking tips, and returns a personalized recipe tailored to the user's request. This project teaches prompt engineering, AI-powered content generation, and conversational workflow design in n8n, making it an excellent foundation for building intelligent food, nutrition, and cooking automation workflows.